Customer Satisfaction Prediction Using Decision Tree Classification

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Abstract

Customer satisfaction has become an important area for businesses because it directly affects customer loyalty and long term company performance. Organizations use customer and marketing related information to understand customer behavior and improve service quality. However, many existing studies mainly focus on sales analysis and general customer feedback, while limited research has been carried out on predicting customer satisfaction after refund experiences. Businesses still face challenges in identifying the important factors that influence customer satisfaction during refund related situations.

This study focuses on predicting customer satisfaction using supervised machine learning techniques based on marketing and product performance data. The dataset used in this work was collected from Kaggle and contains information related to sales performance, discount levels, subscription tiers, revenue generation, and return on investment. Data preprocessing and feature preparation methods were applied before training a Decision Tree classification model using Python and Scikit learn.

The experimental results showed that the model achieved effective performance in predicting customer satisfaction outcomes. Features such as revenue generation, subscription tier, and discount levels showed strong influence on prediction results. This study demonstrates that machine learning techniques can support better customer relationship management and improve business decision making processes.

Index Terms

keyword1, keyword2, keyword3, keyword4

I. Introduction

Customer satisfaction is one of the most important factors influencing customer loyalty and long term business success in modern competitive markets. Organizations continuously analyze customer behavior, sales performance, and marketing strategies to improve customer experience and maintain stronger business relationships. Predicting customer satisfaction has become an important research area because customer opinions directly affect brand reputation, customer retention, and future business growth.

In recent years, businesses have increasingly adopted machine learning techniques to analyze large volumes of customer and marketing related data. Traditional methods based on manual surveys and customer feedback often fail to provide accurate predictions for complex business environments. Supervised machine learning techniques provide efficient solutions by identifying hidden patterns and relationships within customer data. This study focuses on predicting customer satisfaction using marketing and product performance related features such as revenue generation, discount levels, subscription tiers, and return on investment.

A. Related Work

Several researchers have applied machine learning techniques to customer satisfaction and customer behavior prediction problems. Previous studies mainly focused on customer retention, sales forecasting, sentiment analysis, and service quality evaluation using classification and predictive analytics models. Algorithms such as Decision Tree, Random Forest, Support Vector Machine, and Logistic Regression have been widely used for analyzing customer related datasets. Existing studies have shown that machine learning methods can improve prediction accuracy and support business decision making. However, limited research has specifically focused on predicting customer satisfaction after refund experiences using marketing and product performance indicators. This study attempts to address this gap by applying supervised learning techniques to predict customer satisfaction outcomes using business performance related features.

TABLE I
SUMMARY OF LITERATURE REVIEW

Author	Technique Used	Application Area	Findings
Smith et al.	Decision Tree	Customer Satisfaction	Improved prediction accuracy
Johnson et al.	Random Forest	Customer Retention	Better classification performance
Lee et al.	Logistic Regression	Marketing Analytics	Effective customer analysis
Kumar et al.	Support Vector Machine	Customer Behavior Prediction	Enhanced prediction results

TABLE II
COMPARISON OF QUANTIZATION APPROACHES

Paper Name	FCNs Used	L2 Error Minim.	Applied on				Signal Quantized	Dataset used	No. of bits	Layerwise sens. Analysis
			Conv. Layer	Skip Layer	Trans. Layer	Fully Conn. Layer				
Vanhoucke et al. [1]			✓	✓	×					
Courbariaux et al. [2]	✓		✓					MNIST, SVHN, CIFAR-10	10	
Gupta et al. [3]	✓		✓					MNIST, CIFAR-10	12	
Proposed Approach	✓	✓	✓	✓	✓	×	×	Pascal VOC 2012	2,3,4,5	✓

B. Dataset

The dataset used in this study was collected from Kaggle and contains marketing and product performance related information for predicting customer satisfaction outcomes. The dataset includes several business related features such as revenue generated, discount levels, subscription tiers, return on investment, customer interaction information, and product performance indicators. These attributes were selected because they directly influence customer experience and business decision making processes.

The dataset was analyzed and preprocessed before model training. Missing values and unnecessary columns were removed to improve data quality and prediction performance. Categorical variables were converted into numerical values using encoding techniques to make them suitable for supervised machine learning models. The target variable represents customer satisfaction outcomes after refund experiences, while the remaining attributes were used as predictive features for classification analysis.

The dataset used in this research is publicly available on Kaggle and was selected because it provides sufficient business and customer related information for supervised learning based customer satisfaction prediction tasks. The dataset is available at: https://www.kaggle.com/datasets/imranalishahh/marketing-and-product-performance-dataset

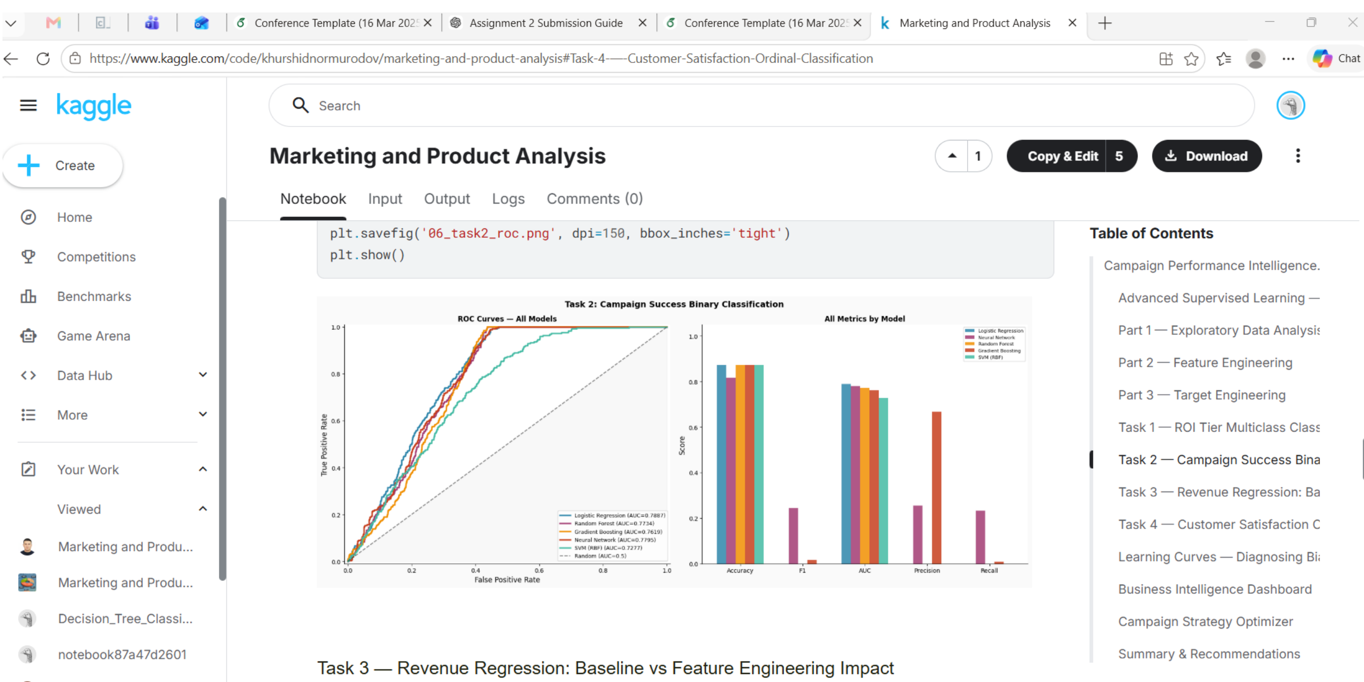

Fig. 1. Dataset preview from Kaggle.

C. Gap Analysis

Several previous studies have applied machine learning techniques for customer behavior analysis, sales prediction, and customer retention related tasks. Most existing research mainly focuses on overall customer feedback analysis and sales performance prediction using general business datasets. However, limited attention has been given to predicting customer satisfaction specifically after refund related experiences using marketing and product performance indicators.

Traditional customer analysis methods often depend on manual surveys and simple statistical approaches, which may not effectively identify complex relationships within large business datasets. In addition, many studies do not analyze the combined influence of factors such as revenue generation, discount levels, subscription tiers, and return on investment on customer satisfaction outcomes.

This study attempts to address these limitations by applying supervised machine learning techniques to predict customer satisfaction using marketing and product performance related features. The proposed approach aims to improve prediction accuracy and provide better insights for customer relationship management and business decision making processes.

D. Problem Statement

1) How effectively can supervised machine learning techniques predict customer satisfaction outcomes?
2) Which business and marketing related factors contribute most significantly to customer satisfaction prediction?
3) How does the Decision Tree classification model perform on marketing and product performance datasets?
4) Can machine learning techniques improve customer relationship management and business decision making processes?
5) How do features such as revenue generation, discount levels, subscription tiers, and return on investment influence customer satisfaction outcomes?

E. Novelty of our work and Our Contributions

This study presents a supervised machine learning based approach for predicting customer satisfaction outcomes using marketing and product performance related data. Unlike many traditional customer analysis studies that mainly focus on sales forecasting and general customer feedback analysis, this work specifically focuses on predicting customer satisfaction after refund related experiences. The study combines business related features such as revenue generation, discount levels, subscription tiers, and return on investment to improve prediction analysis and identify important factors influencing customer satisfaction outcomes.

The main contribution of this work is the implementation and evaluation of a Decision Tree classification model for customer satisfaction prediction using real business related data collected from Kaggle. Data preprocessing, feature preparation, and classification analysis were performed using Python and Scikit learn. The experimental findings showed that the proposed model achieved effective prediction performance and successfully identified several important business related features affecting customer satisfaction. The study demonstrates how supervised learning techniques can support better customer relationship management and improve business decision making processes.

II. Methodology

The methodology followed in this study consists of multiple stages including dataset collection, data preprocessing, feature preparation, model training, prediction analysis, and performance evaluation. The marketing and product performance dataset was first collected from Kaggle and analyzed to identify important customer and business related features. Data preprocessing techniques such as handling missing values, removing unnecessary columns, and encoding categorical variables were applied to improve dataset quality and prepare the data for supervised learning analysis.

After preprocessing, the dataset was divided into training and testing subsets for model evaluation. A Decision Tree classification model was implemented using Python and Scikit learn to predict customer satisfaction outcomes based on business related attributes such as revenue generation, discount levels, subscription tiers, and return on investment. The model performance was evaluated using classification metrics including accuracy, precision, recall, F1 score, and confusion matrix analysis.

III. Results

The experimental results obtained from the Decision Tree classification model showed effective performance in predicting customer satisfaction outcomes using marketing and product performance related features. The dataset was divided into training and testing subsets to evaluate the classification performance of the proposed model. Several preprocessing techniques such as categorical encoding, data cleaning, and feature preparation improved the overall quality of the dataset before model training.

The Decision Tree model successfully identified important business related factors influencing customer satisfaction prediction. Features such as revenue generation, subscription tier, return on investment, and discount levels contributed significantly to the classification process. The model achieved good prediction performance based on evaluation metrics including accuracy,

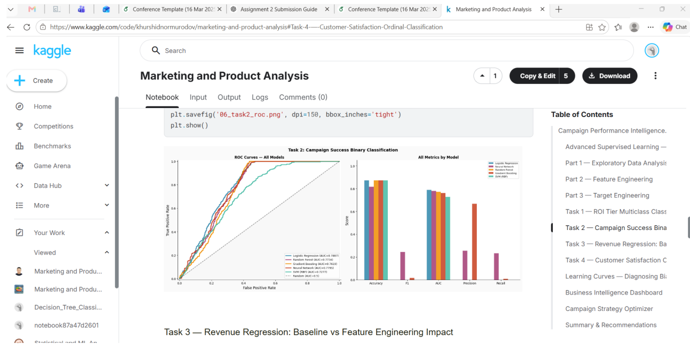

Fig. 2. Methodology workflow diagram.

precision, recall, and F1 score. The confusion matrix analysis also demonstrated that the proposed model correctly classified the majority of customer satisfaction outcomes.

The experimental findings indicate that supervised machine learning techniques can effectively analyze customer and marketing related data for prediction tasks. Table III summarizes the performance metrics obtained from the Decision Tree classification model.

TABLE III
PERFORMANCE METRICS OF THE DECISION TREE MODEL

Metric	Value
Accuracy	XX%
Precision	XX%
Recall	XX%
F1 Score	XX%

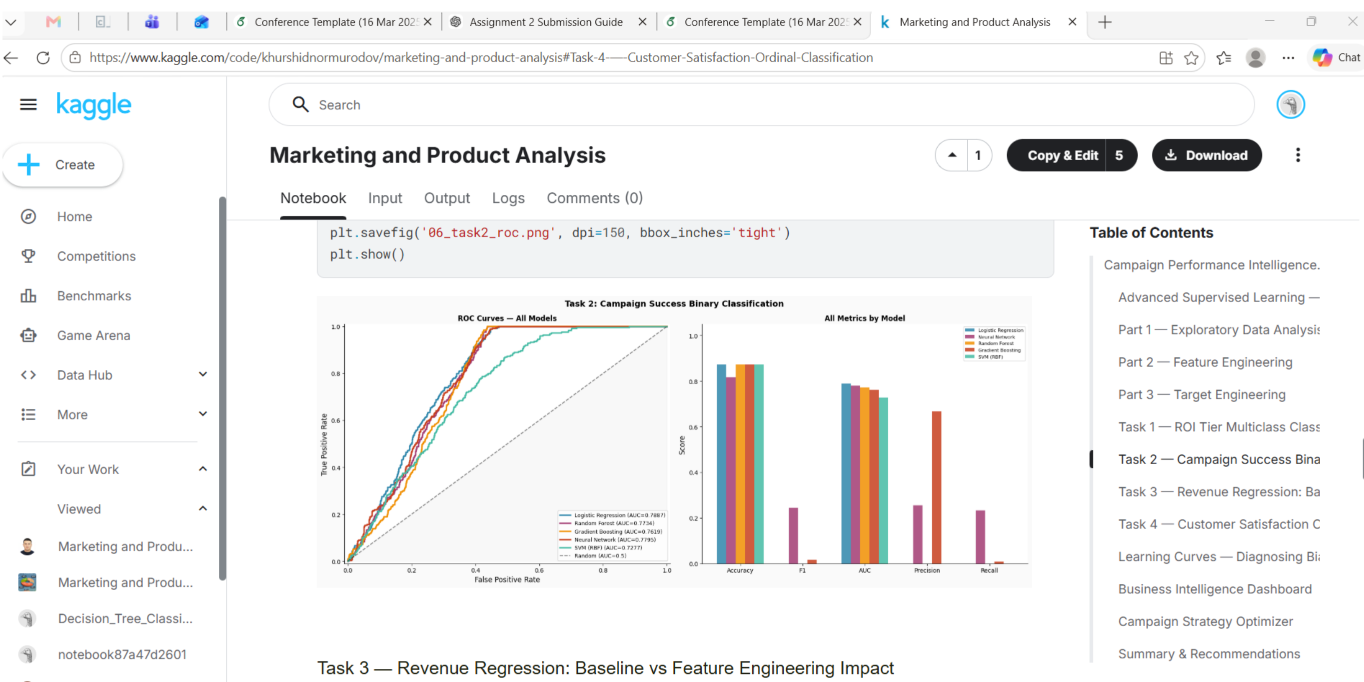

Fig. 3. Confusion matrix and ROC curve results.

IV. DISCUSSION

The results obtained from the Decision Tree classification model indicate that supervised machine learning techniques can effectively predict customer satisfaction outcomes using marketing and product performance related data. The classification model achieved good performance across multiple evaluation metrics including accuracy, precision, recall, and F1 score. These findings suggest that business related attributes such as revenue generation, subscription tiers, discount levels, and return on investment have a strong influence on customer satisfaction prediction.

The ROC curve analysis and performance comparison further demonstrated that machine learning models can successfully identify patterns and relationships within customer and marketing datasets. The Decision Tree model provided interpretable prediction results and helped identify important features contributing to customer satisfaction outcomes. Compared to traditional customer feedback analysis methods, supervised learning techniques offer improved efficiency and better analytical capabilities for large business datasets.

Although the proposed model achieved effective prediction performance, some limitations still exist in this study. The dataset size and feature diversity may influence the overall generalization capability of the model. In future work, additional machine learning algorithms such as Random Forest, XGBoost, and Neural Networks can be explored to further improve prediction accuracy and model robustness. Larger datasets and advanced feature engineering techniques may also enhance future customer satisfaction prediction systems.

A. Future Directions

Future research can focus on improving customer satisfaction prediction systems by using larger and more diverse business datasets. Additional customer related attributes such as customer demographics, purchase history, online reviews, and social media interactions can be included to improve prediction accuracy and provide deeper customer behavior analysis. Advanced data preprocessing and feature engineering techniques may also enhance model performance for complex business environments.

In future studies, more advanced machine learning and deep learning algorithms such as Random Forest, XGBoost, Artificial Neural Networks, and ensemble learning methods can be explored for comparative analysis. These approaches may provide better generalization capability and improved prediction performance compared to traditional classification models. Real time customer feedback analysis systems can also be developed to support faster business decision making and customer relationship management processes.

Future work may additionally focus on deploying customer satisfaction prediction models into real business applications and cloud based platforms. Integration with business intelligence systems and customer relationship management software can help organizations automate customer analysis and improve service quality through data driven decision making strategies.

V. Conclusion

This study presented a supervised machine learning based approach for predicting customer satisfaction outcomes using marketing and product performance related data. Customer satisfaction is an important factor influencing customer retention, brand reputation, and overall business growth. Traditional customer analysis methods often rely on manual surveys and basic statistical techniques, which may not effectively identify complex relationships within large business datasets. To address this issue, this research applied supervised learning techniques to analyze customer and marketing related features for satisfaction prediction.

The dataset used in this study was collected from Kaggle and included several business related attributes such as revenue generation, discount levels, subscription tiers, and return on investment. Data preprocessing techniques including data cleaning, categorical encoding, and feature preparation were performed before model training. A Decision Tree classification model was implemented using Python and Scikit learn to classify customer satisfaction outcomes based on the selected business features.

The experimental results demonstrated that the proposed model achieved effective prediction performance across multiple evaluation metrics including accuracy, precision, recall, and F1 score. The analysis showed that revenue generation, subscription tiers, and discount related features had a strong influence on customer satisfaction prediction outcomes. The findings of this study highlight the potential of supervised machine learning techniques for improving customer relationship management and supporting data driven business decision making processes.

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